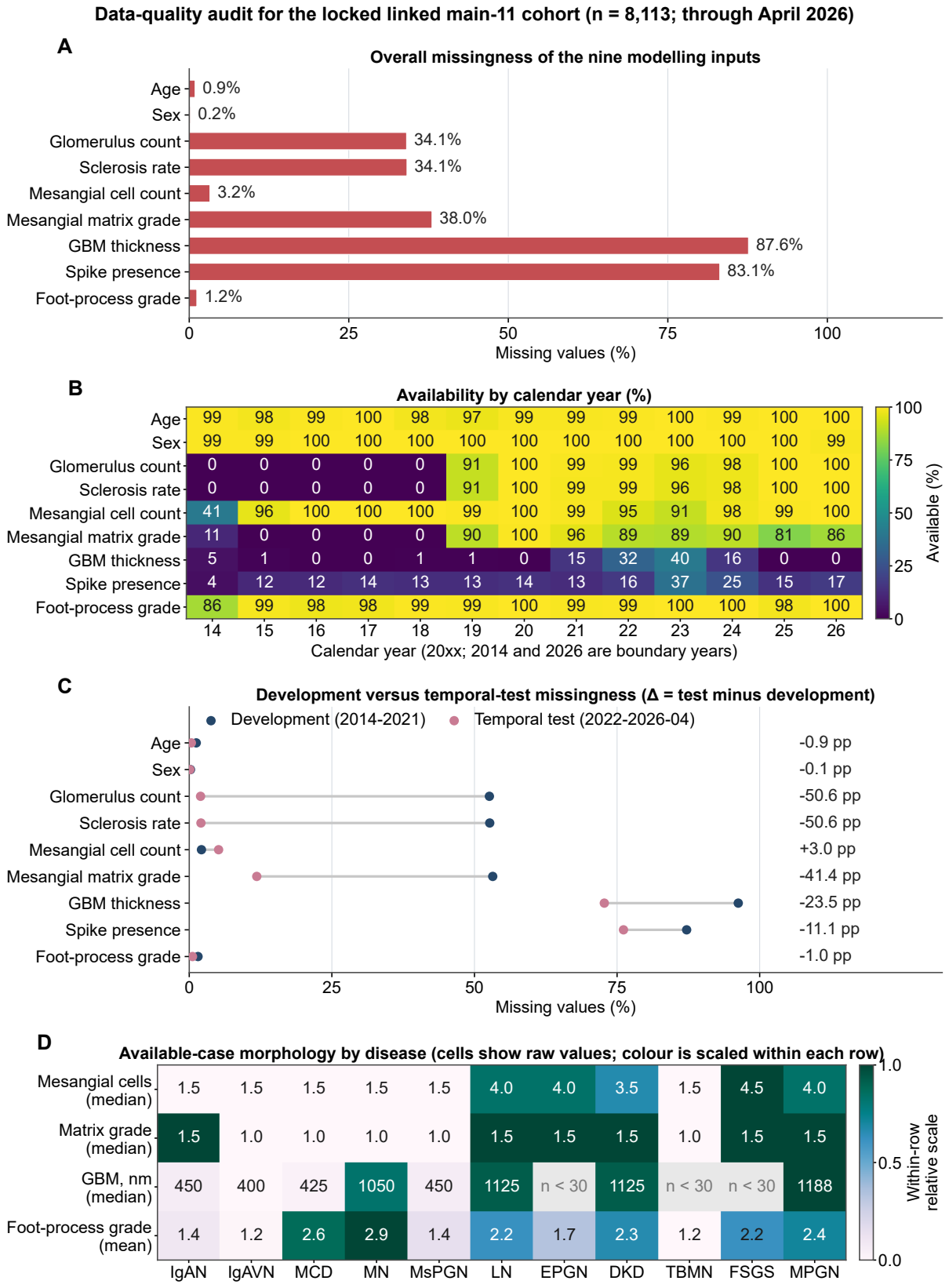

Data-quality audit for the locked linked main-11 cohort (n = 8,113; through April 2026)
A
Overall missingness of the nine modelling inputs
Age
Sex
Glomerulus count
Sclerosis rate
Mesangial cell count
Mesangial matrix grade
GBM thickness
Spike presence
Foot-process grade
0.9%
0.2%
34.1%
34.1%
3.2%
38.0%
87.6%
83.1%
1.2%
0
25
50
75
100
Missing values (%)
B
Availability by calendar year (%)
Age 99 98 99 100 98 97 99 99 99 100 99 100 100
Sex 99 99 100 100 100 100 100 100 100 100 100 100 99
Glomerulus count 0 0 0 0 0 91 100 99 99 96 98 100 100
Sclerosis rate 0 0 0 0 0 91 100 99 99 96 98 100 100
Mesangial cell count 41 96 100 100 100 99 100 99 95 91 98 99 100
Mesangial matrix grade 11 0 0 0 0 90 100 96 89 89 90 81 86
GBM thickness 5 1 0 0 1 1 0 15 32 40 16 0 0
Spike presence 4 12 12 14 13 13 14 13 16 37 25 15 17
Foot-process grade 86 99 98 98 99 99 100 99 99 100 100 98 100
14 15 16 17 18 19 20 21 22 23 24 25 26
Calendar year (20xx; 2014 and 2026 are boundary years)
100
75
50
25
0
Available (%)
C
Development versus temporal-test missingness (Δ = test minus development)
Development (2014-2021)
Temporal test (2022-2026-04)
Age -0.9 pp
Sex -0.1 pp
Glomerulus count -50.6 pp
Sclerosis rate -50.6 pp
Mesangial cell count +3.0 pp
Mesangial matrix grade -41.4 pp
GBM thickness -23.5 pp
Spike presence -11.1 pp
Foot-process grade -1.0 pp
0
25
50
75
100
Missing values (%)
D
Available-case morphology by disease (cells show raw values; colour is scaled within each row)
Mesangial cells (median) 1.5 1.5 1.5 1.5 1.5 4.0 4.0 3.5 1.5 4.5 4.0
Matrix grade (median) 1.5 1.0 1.0 1.0 1.0 1.5 1.5 1.5 1.0 1.5 1.5
GBM, nm (median) 450 400 425 1050 450 1125 n < 30 1125 n < 30 n < 30 1188
Foot-process grade (mean) 1.4 1.2 2.6 2.9 1.4 2.2 1.7 2.3 1.2 2.2 2.4
IgAN IgAVN MCD MN MsPGN LN EPGN DKD TBMN FSGS MPGN
1.0
0.5
0.0
Within-row relative scale